

4.3.3 Yield and atom economy of chemical reactions (chemistry only)

4.3.3.1 Percentage yield

Content	Key opportunities for skills development
Even though no atoms are gained or lost in a chemical reaction, it is not always possible to obtain the calculated amount of a product because: - the reaction may not go to completion because it is reversible - some of the product may be lost when it is separated from the reaction mixture - some of the reactants may react in ways different to the expected reaction. The amount of a product obtained is known as the yield. When compared with the maximum theoretical amount as a percentage, it is called the percentage yield. $\% \text{ Yield} = \frac{\text{Mass of product actually made}}{\text{Maximum theoretical mass of product}} \times 100$ Students should be able to: - calculate the percentage yield of a product from the actual yield of a reaction (HT only) calculate the theoretical mass of a product from a given mass of reactant and the balanced equation for the reaction.	WS 4.2, 4.6 MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages. MS 2a Use an appropriate number of significant figures. MS 3b Change the subject of an equation.

4.3.3.2 Atom economy

Content	Key opportunities for skills development
The atom economy (atom utilisation) is a measure of the amount of starting materials that end up as useful products. It is important for sustainable development and for economic reasons to use reactions with high atom economy. The percentage atom economy of a reaction is calculated using the balanced equation for the reaction as follows: $\frac{\text{Relative formula mass of desired product from equation}}{\text{Sum of relative formula masses of all reactants from equation}} \times 100$ Students should be able to: - calculate the atom economy of a reaction to form a desired product from the balanced equation (HT only) explain why a particular reaction pathway is chosen to produce a specified product given appropriate data such as atom economy (if not calculated), yield, rate, equilibrium position and usefulness of by-products.	WS 4.2, 4.6 MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages. MS 3b Change the subject of an equation.